3.1

- (a) Standard error measures the variability of the estimator, that is how much the estimate changes when taking repeated samples. Standard deviation measures the variability of the data in the population. Neither can be observed and only be estimated.
- (b) Under the sharp null, Y(1)=Y(0) for all units. Thus, randomisation inference fills the missing potential outcomes with the same observed values, simulate all possible treatment assignments or a large random sample of them, calculate the ATE estimates for the different samples, calculate the proportion of the estimates sitting below the original estimate, and take that as the p value.
- (c) A 95% confidence interval is an interval that has a 95% probability of containing the true population parameter.
- (d) Complete random assignment pools all subjects together and assigns them to treatment/control according to appropriate procedures. Block random assignment partitions participants into subgroups and randomises treatment assignment within those groups. This procedure reduces sampling variability by grouping subjects with similar potential outcomes. Cluster random assignment assigns treatment/control to groups of subjects.
- (e) Balanced design minimises sampling variability under the assumption that treatment and control groups have similar variances.

3.5

- (a) Village 1 treated: -0.83; village 2 treated: 0; village 3 treated: 15.83; village 4 treated: 0.83; village 5 treated: 4.17; village 6 treated: 0; village 7 treated: 15
- (b) average of estimates equals true ATE